

Question:

Benzodiazepines are clinically useful because of their inhibitory effects on the central nervous system. Which of the following correctly pairs the site of action of benzodiazepines with the molecular mechanism by which they exert their effects? Answer Choices: A. GABA-A receptors; blocking action of GABA B. GABA-B receptors; activating a G-protein coupled receptor C. GABA-A receptors; increasing the duration of activation of a chloride ion channel D. GABA-A receptors; increasing the frequency of activation of a chloride ion channel

Reasoning + Answer:

<think> Given that benzodiazepines notably act on GABA-A receptors to enhance GABA's effects, their capability to increase chloride channel activation frequency aligns with defining their therapeutic endpoint effectively. </think> <answer> GABA-A receptors; increasing the frequency of activation of a chloride ion channel </answer>

